Lab 03

Visualizing changing Canadian environments

Teaching Week 03 | Individual work. Submit in R or Python.

Purpose

Use three complementary graphics to answer one question about a Canadian environmental measure. Prepare the data with the Week 02 workflow, then choose scales and comparisons that help a reader interpret its variation.

Learning outcomes

After completing this lab, you should be able to

- match chart form to variable type and analytical question;
- encode comparisons with honest scales and accessible design;
- compare distributions and groups with documented bins, bandwidths, and quartiles; and
- connect graphical evidence to a bounded written claim.

Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

Option A | Canada's Air Pollutant Emissions Inventory

Publisher. Environment and Climate Change Canada
Official record. https://open.canada.ca/data/en/dataset/fa1c88a8-bf78-4fcb-9c1e-2a5534b92131
Record identifier. `fa1c88a8-bf78-4fcb-9c1e-2a5534b92131`
Licence and format. Open Government Licence – Canada; ECCC Data Mart with delimited data files
Expected variables. year; province; sector; pollutant; emissions
Use in this lab. Visualize change by place, sector, or pollutant.
Why it fits. Several dimensions support honest comparisons and alternative chart forms.

Download `EN_APEI-Can-Prov_Terr.csv` from ECCC Data Mart's `A-APEI_Tables_Canada_Provinces_Territories` folder using its download control, not by saving the page. Select one pollutant, column-header unit, inventory edition, and sector level. Annual emission estimates are not ambient concentrations. Keep national, provincial, and sector totals separate from their components, and record the estimates and years retained.

Option B | Forest management and disturbances

Publisher. Environment and Climate Change Canada
Official record. https://open.canada.ca/data/en/dataset/426704e8-5a25-49fc-8a77-d80481a08460
Record identifier. `426704e8-5a25-49fc-8a77-d80481a08460`
Licence and format. Open Government Licence – Canada; CSV
Expected variables. year; region; activity; disturbance; area
Use in this lab. Visualize land disturbance patterns.
Why it fits. Temporal and regional measures support trends, distributions, and composition.

The resource *Area burned by forest fires by region* gives annual regional values in square kilometres, with title lines, national and regional figures, and notes. Document the edition and time coverage you retrieve, and justify your handling of aggregate and regional values. Read the region definitions and the Nunavut coverage limitation first.

Option C | Municipal Solid Waste Curbside Collection Tonnage

Publisher. Nova Scotia Environment and Climate Change
Official record. https://data.novascotia.ca/d/r7ci-8szi
Record identifier. `r7ci-8szi`
Licence and format. Nova Scotia Open Government Licence; CSV
Expected variables. year; municipality; region; waste stream tonnage; service units
Use in this lab. Visualize waste change across regions.
Why it fits. Clear measures and groups support focused statistical graphics.

Export CSV from the catalogue page. Each row is a municipality and reporting year whose label spans two calendar years, with garbage, recyclables, organics, and totals in tonnes, service-unit counts, and published per-unit fields. A missing report is not a measured zero, and the publisher notes that blanks or zeros in this file mean no data was available. Explain how units, normalization, aggregate records, and unavailable denominators affect the comparison.

Data contract

Choose one listed dataset and define a focused visual question about environmental change.

- Retain at least 100 observations across at least three groups or five time periods.
- Include one numeric measure and one categorical, temporal, or regional comparison variable.
- Use Week 01 summaries and Week 02 cleaning before plotting.
- Produce at least three complementary statistical graphics.
- Use honest scales and show observed variation. If the publisher supplies uncertainty estimates, explain them before plotting them. No confidence interval or significance test is required.
- Include accessible labels, units, captions, and colour choices.
- Document the official record URL, publisher, licence, retrieval steps, and access date.

What the contract does not settle

Aggregated region-year or municipality-year values describe those reporting units, not individual fires, households, or emission sources.

Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I stated a focused question about change or comparison.
- ☐ My table satisfies every rule in the data contract.
- ☐ I completed summaries and cleaning before plotting.
- ☐ I selected chart forms that match the variables and question.
- ☐ I will show units, variation, accessible labels, and honest scales.
- ☐ I will cite the dataset in the report.

Cumulative method

This lab carries forward Week 01 summaries and outlier investigation together with the Week 02 cleaning pipeline. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

Work program

1. State the visual question

Define the measure, observational unit, comparison, and time or geographic scope in one sentence.

2. Prepare and verify

Reuse the earlier audit and cleaning workflow. Reconcile denominators, units, missing periods, and group definitions before plotting.

3. Build complementary views

Produce a distribution view, a comparison or trend view, and one detailed view that reveals structure hidden by an aggregate. Explain what each plotted value represents. For example, a distribution of municipality-year tonnages describes variation among those totals, not variation among households. Use facets or a second grouping so the third view adds something the first two do not.

4. Audit graphical integrity

Explain scale limits, transformations, ordering, colour, and any aggregation. Keep missing periods visible instead of replacing them with zero. If you show uncertainty, identify its source and meaning. Identify one misleading alternative and why you rejected it.

5. Write the visual argument

Link each figure to one evidence-based claim and end with a limitation that a reader could not infer from the graphics alone. Save the figures under `results/` and include them in the report. Use a distinct filename for each figure.

Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Downloaded data and generated results are not course source, so they never go into the course repository. Commit the report, its source, and the generated tables and figures it cites to your private work repository. Keep that evidence private until the deadline, and publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-03/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on October 1, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-03/`. A branch URL or movable tag alone is not a fixed submission. The package includes

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, figures, and any summary statistics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

Assessment standard

The course uses one rubric across all integrated labs. The best eight of the nine weekly lab results contribute 15% of the course grade.

Table 1. Integrated lab marking criteria.

Criterion	Weight (%)
Dataset provenance and suitability	15
Reuse of earlier methods	20
Current-week analytical method	35
Interpretation and limitations	20
Reproducibility and communication	10
Total	100

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.